

ALEX MORGAN

Staff Distributed Systems & Cloud Infrastructure Engineer

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EXECUTIVE SUMMARY

Staff-level distributed systems architect with 8+ years designing fault-tolerant runtimes, sub-millisecond event streaming pipelines, and high-concurrency storage engines in Rust, Go, and eBPF. Open-source lead.

PROFESSIONAL EXPERIENCE

Staff Infrastructure Engineer - Veloce Systems (San Francisco, CA) **2022 - Present**

Architected distributed control plane managing over 450,000 concurrent tenant workloads with p99 < 1.2ms. Engineered gossip-based failure detector and consensus sharding, slashing cross-zone egress traffic by 38%. Mentored 12 senior/staff engineers and standardized OpenTelemetry tracing across all fleet services.

Senior Backend Engineer - Aether Cloud (Seattle, WA) **2019 - 2022**

Owned high-throughput streaming telemetry pipeline ingesting 1.2 billion events daily with zero data loss. Rewrote CPU-intensive ingestion microservices in Rust, saving \$420,000 annually in AWS compute spend. Built append-only write-ahead log (WAL) engine backed by NVMe storage with strict ACID durability.

Software Engineer - Hexagon Labs (Boston, MA) **2017 - 2019**

Developed customer-facing telemetry visualization dashboards and real-time GraphQL gateway microservices. Implemented smart batching query algorithms, reducing Postgres read amplification by 65%.

KEY PROJECTS & OPEN-SOURCE ARCHITECTURE

KestrelDB: Embedded Vectorized Time-Series Engine [Rust, Apache Arrow, SIMD, LSM-Tree, gRPC]
Open-source columnar database yielding 2M writes/sec on commodity hardware with adaptive Gorilla compression.

SynapseGate: eBPF Kubernetes Ingress Controller [eBPF, Go, Linux Kernel, Kubernetes]
Zero-overhead socket routing directly in kernel space, reducing inter-pod network latency by 45%.

CORE COMPETENCIES & TECHNICAL PROFICIENCIES

Programming Languages Rust, Go, TypeScript, C/C++, Python, SQL SIMD intrinsics, WebAssembly (Wasm)	Systems & Storage Engines Raft/Paxos Consensus, LSM-Trees, eBPF PostgreSQL, Apache Kafka, ClickHouse, Redis
Cloud & DevOps Infrastructure Kubernetes, Docker, Terraform / OpenTofu AWS, GCP, Linux Kernel Tuning, GitHub Actions	Architecture & Observability Distributed Tracing, OpenTelemetry, gRPC Event-Driven SEDA, Microservices, Protobuf

EDUCATION & CERTIFICATIONS

M.S. in Computer Science (Distributed Systems) - Carnegie Mellon University	3.94 GPA / Dean Honors
B.S. in Computer Engineering - University of Michigan	Summa Cum Laude
Certified Kubernetes Administrator (CKA) - Linux Foundation & CNCF (2023)	